Shuo Sun

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Research Interests

Modeling and algorithm design for supply chain and revenue management using optimization and machine learning

Education

University of California, Berkeley *08/2021 – Current*
Ph.D. Candidate in Industrial Engineering and Operations Research
Minor in Statistics & Algorithm Design
Advisors: Zuo-Jun Max Shen & Rajan Udwani

Tsinghua University *08/2016 – 06/2021*
B.E. in Industrial Engineering

University of Wisconsin-Madison *08/2019 – 12/2019*
Exchange student in the Department of Industrial and Systems Engineering

Industry Experience

Amazon, Seattle, WA *05/2024 – 08/2024*
Applied scientist intern at Fashion, Fitness & Creators

Honors & Awards

Honorable mention, POMS-HK Best Student Paper Competition *2025*

Winner, INFORMS Daniel H. Wagner Prize *2024*

Finalist, INFORMS RMP Jeff McGill Best Student Paper Award *2024*

IEOR Ph.D. First-year Fellowship *2021–2022*

Finalist, INFORMS Undergraduate Operations Research Prize *2020*

Preprints & Publications

[1] "A Markovian Approach for Cross-Category Complementarity in Choice Modeling", with Omar El Housni and Rajan Udwani. Submitted to *Management Science* [link]

[2] "A Unified Algorithmic Framework for Dynamic Assortment Optimization under MNL Choice", with Rajan Udwani and Zuo-Jun Max Shen. Major revision at *Operations Research* [link]

- *Proceedings of the 26th ACM Conference on Economics and Computation* [link]
- **Finalist, 2024 INFORMS RMP Jeff McGill Student Paper Award**
- **Honorable Mention, 2025 POMS-HK Best Student Paper Competition**

[3] "JD.com Improves Fulfillment Efficiency with Data-driven Integrated Assortment Planning and Inventory Allocation", with Zuo-Jun Max Shen, Yongzhi Qi, Hao Hu, Ningxuan Kang, Jianshen Zhang, Xin Wang, Xiaoming Lin. *INFORMS Journal on Applied Analytics* [link]

- **Winner of the 2024 INFORMS Daniel H. Wagner Prize** [link]

[4] "Online MDP with Prototypes Information: A Robust Adaptive Approach", with Meng Qi and Zuo-Jun Max Shen. *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence* [link]

[5] "Robust Home Healthcare Districting and Workforce Management with Crowdsourced Caregivers", with Mingda Liu, Xiaolei Xie, and Zuo-Jun Max Shen. Submitted to *Productions and Operations Management*

Teaching

Graduate Student Instructor, Economics of Supply Chains — 2023 Spring, 2025 Spring

Graduate Student Instructor, Supply Chain & Logistics Management — 2024 Spring

Graduate Student Instructor, Optimization Analytics — 2022 Fall, 2023 Fall

Services

Reviewer for Management Science, Operations Research, Manufacturing and Service Operations Management, MSOM SIG

Conference Talks

"A Markovian Approach for Cross-Category Complementarity in Choice Modeling"

- INFORMS RMP 2025, INFORMS Annual Meeting 2025 (scheduled)

"A Unified Algorithmic Framework for Dynamic Assortment Optimization under MNL Choice"

- POMS-HK 2025, INFORMS RMP 2024, INFORMS Annual Meeting 2023, Purdue Supply Chain and Operations Management Conference 2025

"JD.com Improves Fulfillment Efficiency with Data-driven Integrated Assortment Planning and Inventory Allocation"

- INFORMS Analytics 2024 (award session)

"Online MDP with Prototypes Information: A Robust Adaptive Approach"

- ISMP 2024, AAAI 2025 (poster presentation)

Other Experience

Member, Berkeley IEOR Grad Student Services Leadership Group *2024, 2025*

Panelist, Berkeley IEOR Ph.D. Visiting Day *2023, 2024*

Mentor for new Ph.D. students, Berkeley IEOR *2023*

Panelist, Berkeley IEOR PhD Orientation Day *2022*